

RESCUE-NET

MASTER ARCHITECTURE & DEVELOPMENT HANDBOOK

VOLUME 3

DATA & INTEGRATION ARCHITECTURE

Versi 1.0 | 1 Agustus 2026
Internal Project Reference

Fokus: model data, schema konseptual, data governance, API, event, audit, integrasi, offline sync, migrasi, dan prioritas implementasi.

Kendali Dokumen

Atribut	Nilai
Judul	Rescue-Net Volume 3 - Data & Integration Architecture
Versi	1.0
Tanggal	1 Agustus 2026
Status	Baseline data and integration architecture
Ruang lingkup	Data model, data governance, API, integration, audit, event, sync, migration, performance
Prinsip utama	Raw Report bukan Operational Truth; provenance dan audit trail tidak boleh hilang

Dasar desain

Rescue-Net menyimpan laporan mentah, evidence, proses verifikasi, hasil konsolidasi, dan keputusan operator sebagai lapisan yang berbeda. Tidak ada satu tabel atau badge yang boleh menyederhanakan seluruh proses menjadi sekadar `verified=true`.

Daftar Isi Ringkas

1. 1. Tujuan Arsitektur Data
2. 2. Data Principles
3. 3. Classification dan Ownership
4. 4. Conceptual Data Model
5. 5. Disaster Event dan Incident
6. 6. Raw Report Model
7. 7. Evidence Model
8. 8. Location dan Wilayah
9. 9. Organization, Posko, dan Verifier
10. 10. Verification Model
11. 11. Need dan Unit Normalization
12. 12. Consolidation dan Operational Truth
13. 13. Command Correction
14. 14. Resource dan Distribution
15. 15. War Room Read Models
16. 16. Audit, Lineage, Versioning
17. 17. Database Standards
18. 18. Indexing dan Performance
19. 19. API Architecture
20. 20. Endpoint Inventory
21. 21. Error, Pagination, Idempotency
22. 22. Event Model
23. 23. Offline Sync Protocol
24. 24. External Integration
25. 25. Security dan Privacy Data
26. 26. Backup, Migration, Restore
27. 27. Data Quality dan Governance
28. 28. Frappe Data Boundary
29. 29. Gap Analysis
30. 30. Immediate Next Actions
31. 31. Acceptance Criteria
32. 32. ADR

1. Tujuan Arsitektur Data

Arsitektur data Rescue-Net harus menghasilkan data operasional yang dapat dipercaya tanpa menghapus jejak laporan asli. Struktur data harus mendukung audit, operasi offline, integrasi lintas organisasi, konsolidasi anti-duplikasi, dan scale-up nasional.

- Memisahkan input mentah dari hasil konsolidasi.
- Menjaga sumber, waktu, lokasi, evidence, dan perubahan.
- Mendukung drill-down nasional hingga posko.
- Mencegah double counting dan silent overwrite.
- Memberi kontrak yang konsisten kepada web, PWA, War Room, dan integrasi eksternal.

2. Data Principles

Prinsip	Implikasi
Immutable raw input	Payload asli, unit asli, source text, dan media reference tidak ditimpa.
Provenance first	Setiap hasil harus dapat dilacak ke sumber.
Layered verification	Identity, location, organization, evidence, dan need memiliki status terpisah.
Versioned operational truth	Kebenaran operasional berubah seiring evidence baru; histori dipertahankan.
MAX by default for overlap	SUM hanya setelah beneficiary/area berbeda terkonfirmasi.
Human authority	AI dan rule engine memberi candidate/rekomendasi.
Idempotent writes	Retry jaringan tidak menciptakan data ganda.
Reference data controlled	Wilayah dan kategori memiliki source version.
Least data necessary	Data pribadi dikumpulkan hanya sesuai kebutuhan operasi.

3. Classification dan Ownership

Kelas data	Contoh	Kontrol
Public	Ringkasan bencana, kebutuhan agregat aman	Publikasi terkontrol dan tanpa data pribadi
Internal	Queue verifikasi, konflik, resource status	Akses operator/organisasi
Restricted	Detail posko, stock, assignment	Scope event/region/organization
Confidential	Identitas pelapor, korban, kontak verifier	Masking, retention, audit akses
Highly confidential	Credential, secret, token	Tidak disimpan di repo atau log

Objek	Business owner	Data steward
Disaster event	Command authority	Event admin
Raw report	Reporter/source	Reporting operator
Operational truth	Command center	Consolidation operator
Organization/posko	Organization admin	Platform admin
Verification	Verifier authority	Verification operator
Resource	Resource owner	Logistics operator

4. Conceptual Data Model

DisasterEvent 1---N Incident
 DisasterEvent 1---N RawReport
 RawReport 1---N Evidence
 RawReport 1---N VerificationResult
 RawReport N---N ConsolidationGroup
 ConsolidationGroup 1---N ConsolidatedNeedVersion
 ConsolidatedNeedVersion 1---N CommandCorrection
 Organization 1---N Posko
 Posko 1---N NeedClaim
 Resource 1---N ResourceMovement
 DistributionPlan 1---N Delivery
 All critical entities -> AuditEvent / SourceReference

Catatan

Relasi konseptual di atas bukan schema final. Codex harus membandingkannya dengan tabel aktual sebelum migration dan tidak boleh menghapus tabel existing hanya karena namanya berbeda.

5. Disaster Event dan Incident

Entity	Field minimum	Rule
disaster_events	uuid, name, type, status, start_at, end_at, lead_authority	Status aktif mengendalikan linking dan visibility
event_areas	event_id, admin_code, geometry/reference, severity	Satu event dapat meliputi banyak area
incidents	event_id, category, location, occurred_at, status	Incident adalah kejadian spesifik di dalam event
event_status_history	event_id, from, to, actor, reason, time	Semua perubahan status diaudit

6. Raw Report Model

Kelompok field	Contoh
Identity	report_uuid, client_uuid, event_id, source_type
Original content	original_text, original_payload, original_language
Time	reported_at_device, received_at_server, occurred_at_claim
Location claim	lat, lon, accuracy, admin codes, manual text
Reporter	reporter_id/contact reference, organization/posko claim
Classification	report_type, need category, urgency candidate
State	pending, in_review, duplicate_candidate, rejected, consolidated
Metadata	device, app version, import source, created_by

- Raw report tidak menyimpan operational_verified sebagai satu boolean universal.
- Normalisasi tidak menghapus original_text atau original_unit.
- client_uuid harus unik per client untuk idempotency.
- Import WhatsApp/OCR/API harus mempertahankan raw payload.

7. Evidence Model

Field	Fungsi
evidence_uuid	Identitas global
report_id / action_id	Objek yang didukung
media_type	photo, video, audio, document, gps, witness
storage_key	Lokasi object storage/file
sha256/checksum	Integrity
captured_at / uploaded_at	Waktu capture dan server
lat/lon/accuracy	Metadata lokasi
device_metadata	Sumber teknis
authenticity_status	unknown, reviewed, suspected, accepted
visibility_class	Public/internal/restricted
retention_class	Kebijakan retensi

Chain of custody

Evidence yang dipakai untuk keputusan harus memiliki jejak upload, akses, perubahan status, dan keterkaitan dengan keputusan. File tidak boleh diganti menggunakan nama/path yang sama tanpa versioning.

8. Location dan Wilayah

Tabel/objek	Isi
wilayah_master	Kode dan nama provinsi, kabupaten/kota, kecamatan, desa/kelurahan
wilayah_versions	Source, version, checksum, effective date
location_claims	Lokasi yang diklaim oleh pelapor
location_validations	GPS/map/admin/manual validation result
location_conflicts	Perbedaan GPS, titik peta, dan admin area
spatial_links	Posko/report/evidence ke wilayah/geometry

- Master utama menggunakan referensi resmi Kemendagri/data.go.id.
- BPS digunakan untuk bridging statistik.
- BIG digunakan untuk boundary/reverse checking.
- Offline app menyimpan subset wilayah yang relevan atau cache bertingkat.

9. Organization, Posko, dan Verifier

Entity	Fungsi
organizations	Organisasi legal/komunitas/NGO/perusahaan
organization_relationships	Parent, chapter, partner, sponsor
posko_nodes	Operational node, coverage, location, ownership
verifier_profiles	Profil verifier
trusted_verification_requests	Permintaan verifikasi
verification_endorsements	Endorsement yang dapat dicabut
organization_memberships	User-role-organization-event scope

- Posko bukan sekadar alamat; ia adalah node operasional.
- Organization verified tidak berarti semua report verified.
- Verifier endorsement memiliki scope dan expiry/revocation.

10. Verification Model

Verification dimension	Status contoh
identity	unverified, pending, verified, rejected, expired
location	gps_verified, map_point, admin_area, review, invalid
organization	unverified, endorsed, verified, revoked
evidence	unreviewed, accepted, insufficient, suspected
need	unreviewed, plausible, conflict, consolidated
operational	not_ready, candidate, approved, superseded

```

verification_results
- uuid
- subject_type / subject_id
- dimension
- status
- verifier_type / verifier_id
- method
- confidence
- reason
- evidence_refs
- valid_from / valid_until
- created_at

```

11. Need dan Unit Normalization

Field	Rule
original_unit_label	Wajib dipertahankan
original_quantity	Nilai dari sumber
normalized_unit	Diisi hanya bila conversion diketahui
normalized_quantity	Hasil conversion yang dapat dijelaskan
conversion_method	Reference/operator/rule
conversion_confidence	Tingkat keyakinan
needs_review_unit_conversion	True bila ambigu
beneficiary_scope	Kelompok penerima
coverage_area	Area kebutuhan

Larangan asumsi kaku

Satu dus, kantong plastik, atau karung tidak memiliki nilai baku lintas komoditas. Sistem tidak boleh memaksakan conversion tanpa evidence atau operator confirmation.

12. Consolidation dan Operational Truth

Entity	Tanggung jawab
duplicate_candidates	Pasangan/group kandidat dengan score dan reasons
overlap_assessments	Kesamaan lokasi, need, time, beneficiary, evidence
consolidation_groups	Kelompok report yang dianalisis bersama
consolidation_decisions	merge, split, reject, keep separate, request evidence
consolidated_needs	Current operational representation
consolidated_need_versions	Histori perubahan nilai/status
source_contributions	Kontribusi setiap report/evidence
scenario_values	minimum, optimal, maximum

same location + same category + same beneficiary

-> default MAX credible value

confirmed different beneficiary/area

-> SUM allowed with operator reason

broad aggregate claim

-> assumption/range layer, not override detail

13. Command Correction

Field	Makna
target_type/id	Operational object yang dikoreksi
before_value	Nilai sebelum
after_value	Nilai sesudah
delta	Selisih eksplisit
reason_code	Kategori alasan
reason_text	Penjelasan operator
evidence_refs	Dasar keputusan
actor/authority	Pelaku dan kewenangan
created_at	Waktu
reversal_of	Referensi koreksi yang dibalik

Contoh

Total consolidated 750 dapat dikoreksi menjadi 1.000 dengan manual correction +250. War Room harus memperlihatkan nilai dasar, koreksi, pelaku, dan alasan; bukan hanya total akhir 1.000.

14. Resource dan Distribution

Entity	Field kunci
resources	type, owner, capacity, availability, location
resource_inventory	quantity, unit, batch, expiry, storage
resource_movements	from, to, qty, status, timestamps
distribution_plans	need refs, priority, route, ETA, authority
deliveries	dispatch, arrival, receipt, evidence
assignments	volunteer/vehicle/team, task, status
delivery_feedback	recipient confirmation, shortfall, issue

15. War Room Read Models

Read model	Optimasi
national_rollup	Hierarchical aggregation dan scenario
event_summary	Critical needs, resources, conflict count
geographic_need_rollup	Province-city-district-village-posko
verification_queue	Dimension, age, priority, assignee
conflict_queue	Duplicate/overlap reason dan impact
resource_coverage	Available vs need vs assigned
command_trace	Decision and correction chronology
sync_health	Offline queue and failed sync trends

- Read model dapat di-rebuild dari transactional data.
- Jangan menjadikan cache/read model sebagai satu-satunya sumber audit.
- Setiap angka War Room harus memiliki drill-down ke source contribution.

16. Audit, Lineage, dan Versioning

Audit element	Isi
actor	User/service/anonymous device
action	create/update/status/merge/split/correct/export
subject	Entity dan ID
before/after	Perubahan relevan
reason	Mandatory untuk action kritis
request_id	Trace teknis
event_id	Scope bencana
time	Server timestamp
source/ip/device	Sesuai privacy policy

Operational Truth

- > ConsolidatedNeedVersion
- > ConsolidationDecision
- > SourceContribution
- > VerificationResult
- > Evidence
- > RawReport

17. Database Standards

Area	Standard
Identifier	UUID untuk business identity; surrogate integer boleh internal
Naming	snake_case, singular/plural konsisten
Time	timestampz untuk waktu operasional
Soft state	status/archived_at; raw tidak hard delete
Foreign key	Gunakan bila lifecycle dan partitioning memungkinkan
JSONB	Untuk raw payload/extension, bukan menggantikan schema inti
Money/quantity	Numeric, bukan float
Coordinates	PostGIS geometry/geography atau numeric tervalidasi
Migration	Forward, tested, backup, rollback/compensating plan

18. Indexing dan Performance

Query	Index/strategi
Active events	status + start/end
Recent reports	event_id + received_at
Location drill-down	admin codes + spatial index
Verification queue	dimension + status + created_at
Duplicate candidate	event/category/time/location signature
Consolidated need	event + area + category + status
Idempotency	unique client_uuid/source key
Audit	subject_type/id + created_at
War Room	Materialized/read model refresh strategy

Index discipline

Index dibuat berdasarkan query nyata dan EXPLAIN, bukan semua kolom. Codex harus memonitor write cost dan ukuran index.

19. API Architecture

- REST untuk command/query umum.
- SSE/WebSocket hanya untuk update War Room yang membutuhkan push.
- Async job untuk AI/media/import besar.
- Endpoint publik dan operator dipisahkan secara permission, bukan logika duplikat.
- OpenAPI harus mencerminkan contract aktual.

Contract	Ketentuan
Versioning	/api/v1 atau header policy yang konsisten
Identifier	UUID string
Time	ISO 8601 dengan timezone
Errors	code, message, details, request_id
Writes	Idempotency-Key untuk operasi retry-prone
Pagination	Cursor untuk data besar
Filtering	Explicit allow-list
Authorization	Role + event + region + organization scope

20. Endpoint Inventory

Domain	Existing/known examples	Target additions
Health	/health	/ready, internal DB/storage checks
Disaster	/disasters	event area/status history
Consolidation	/data-consolidation/summary, /national-rollup	conflict detail, source trace, rebuild status
Command	/command-corrections	reversal, approval, history
Verifier	/public/verifier-profiles, /verification-requests, endorsements	expiry, scope, audit export
Offline	device registrations	batch sync, queue ack, conflict response
Reporting	public report flow	idempotent report/media endpoints
War Room	scenario buttons/read views	drill-down and realtime feed

21. Error, Pagination, dan Idempotency

```
{
  "error": {
    "code": "RN_CONFLICT_REVIEW_REQUIRED",
    "message": "Data memerlukan review operator",
    "details": {"conflict_id": "..."},
    "request_id": "..."}
}
```

HTTP	Penggunaan
400	Payload invalid
401	Authentication required
403	Scope/role denied
404	Resource absent
409	Conflict/idempotency/state transition
422	Semantically invalid
429	Rate limited
500	Unexpected internal error
503	Dependency unavailable/readiness failure

22. Event Model

Event	Payload minimum
RawReportCreated	report_id, event_id, source, time
EvidenceAttached	evidence_id, subject_id, checksum
VerificationChanged	subject, dimension, old/new, actor
DuplicateCandidateCreated	candidate ids, score, reasons
ConsolidationDecided	group, decision, source refs
OperationalTruthUpdated	object, version, old/new summary
CommandCorrectionApplied	target, delta, actor, reason
DistributionStatusChanged	delivery, old/new, evidence
OfflineSyncConflict	client item, server item, conflict fields

- Event bukan pengganti database transaksi pada tahap awal.
- Gunakan outbox pattern sebelum message broker kompleks.
- Event harus memiliki event_id, occurred_at, producer, schema_version, correlation_id.

23. Offline Sync Protocol

Tahap	Perilaku
Capture	Simpan lokal dengan client_uuid
Queue	queued dengan dependency attachment
Handshake	Kirim device/app/user/event context
Batch write	Server menjawab accepted/duplicate/conflict/rejected
Attachment	Checksum dan resumable/retry
Conflict	Tampilkan field dan opsi operator/user
Ack	Client menandai synced hanya setelah server ack
Retry	Backoff + jitter; tidak membuat record baru

Client item status:

draft -> queued -> syncing -> synced

-> retry_wait

-> conflict

-> rejected

24. External Integration

Integrasi	Pattern
Kemendagri/data.go.id	Versioned import/cache
BPS bridging	Reference mapping
BIG boundary	GIS import/service
BMKG/PVMBG	Observation feed dengan source/time
Messaging	Notification adapter
NGO/government	API contract + organization scope
Map/geocoder	Provider abstraction dan cache
Frappe	API/event boundary, bukan direct shared-table coupling

25. Security dan Privacy Data

- Pisahkan PII reporter/korban dari data publik.
- Masking pada UI dan export sesuai role.
- Log tidak boleh memuat credential atau full sensitive payload.
- Evidence access memakai authorization dan expiring link bila object storage.
- Retensi ditetapkan per class; raw audit penting tidak berarti semua PII disimpan selamanya.
- Export data kritis diaudit.

26. Backup, Migration, dan Restore

33. Backup database dan media sebelum migration.
34. Catat schema version dan application image.
35. Jalankan migration pada staging/copy.
36. Uji query utama, consolidation, verifier, dan War Room.
37. Deploy dengan backward compatibility.
38. Jalankan functional health, bukan hanya process health.
39. Siapkan compensating/rollback plan.
40. Lakukan restore drill terpisah.

Larangan

Jangan recreate postgres-main, menghapus volume, reset schema, atau menjalankan migration destructive hanya untuk menghilangkan error startup.

27. Data Quality dan Governance

Dimension	Contoh metric
Completeness	Persentase report dengan lokasi/evidence/category
Validity	Admin code dan unit sesuai reference
Uniqueness	Duplicate confirmed per total report
Timeliness	Report-to-review latency
Consistency	Mismatch location/admin/organization
Traceability	Operational objects dengan full source lineage
Accuracy proxy	Correction/reversal rate
Freshness	Age read model dan master data

28. Frappe Data Boundary

Frappe candidate	Rescue-Net custom authority
User/role administration	Event-scoped operational authorization
Organization master	Operational relationship and trust semantics
Posko form/workflow	Location and consolidation behavior
Approval tasks	Operational Truth decision engine
Administrative reports	War Room read models
Document/audit UI	Immutable raw/evidence provenance

- Gunakan stable UUID lintas sistem.
- Tetapkan system-of-record per entity.
- Hindari two-way sync tanpa conflict policy.
- Jangan berbagi database table secara langsung sebagai pola utama.

29. Gap Analysis

Area	Known baseline	Target	Priority
Schema inventory	Belum terdokumentasi lengkap	Actual ERD + dictionary	P0
Migration history	Perlu audit	Reproducible and tested	P0
DB readiness	Health umum	Functional dependency check	P0
Idempotent sync	Flow konsep/existing partial	Formal contract	P1
Source lineage	Prinsip ada	UI/API trace end-to-end	P1
Wilayah	Plan source	Versioned master/cache	P1
Read models	Summary/rollup ada	Drill-down and refresh control	P1
Privacy/retention	Belum lengkap	Policy and enforcement	P2
Frappe boundary	Konsep hybrid	Pilot contract	P2

30. Immediate Next Actions

30.1 P0 - Audit Existing Data Layer

41. Dump schema-only dan daftar tabel dari rescuenet_db tanpa mempublikasikan credential.
42. Petakan tabel existing terhadap conceptual model volume ini.
43. Audit migration files/history dan startup migration behavior.
44. Verifikasi unique/idempotency constraints.
45. Verifikasi foreign keys, indexes, timestamps, and audit columns.
46. Test backup and restore pada environment terpisah.

30.2 P1 - Tutup Gap Kritis

47. Tambahkan readiness DB dan functional query.
48. Implementasikan official wilayah master/version.
49. Formalize offline sync response statuses.
50. Tambahkan source contribution/trace untuk consolidation.
51. Bangun War Room drill-down yang menjelaskan angka.
52. Tambah data quality metrics.

31. Acceptance Criteria

- Tidak ada raw payload yang hilang saat normalisasi.
- Retry report/sync tidak membuat duplikat.
- Setiap consolidated value dapat ditelusuri ke source.
- MAX/SUM decision tercatat dan dapat dijelaskan.
- Badge verification tidak menyiratkan scope lain.
- Command correction memiliki before/after/reason/actor.
- Backup dapat direstore.
- API errors dan pagination konsisten.
- Wilayah memiliki source_version.
- Frappe pilot tidak mengambil alih consolidation authority.

32. Architecture Decision Records

ADR	Keputusan
ADR-017	Immutable raw report and evidence references
ADR-018	Layered verification model
ADR-019	Versioned consolidated operational objects
ADR-020	UUID and idempotency for offline writes
ADR-021	Official wilayah master with source version
ADR-022	Read models are rebuildable
ADR-023	Outbox before complex event broker
ADR-024	Frappe integration through explicit boundary
ADR-025	No destructive migration without tested restore
ADR-026	Source lineage required for War Room numbers

Lampiran A - Data Dictionary Template

Kolom	Isi
Entity/table	Nama
Business definition	Makna bisnis
System of record	Otoritas
Primary identifier	UUID/key
Fields	Nama, type, nullability
Validation	Rule
PII class	Classification
Retention	Policy
Indexes	Query rationale
Source/lineage	Asal dan downstream
Owner/steward	Penanggung jawab

Lampiran B - Audit Query Checklist

- List tables, views, extensions, schemas.
- List migration/version table.
- List FK and orphan risks.
- List unique constraints and duplicate keys.
- List indexes and unused/large indexes.
- List timestamp columns without timezone.
- List JSONB fields used for core business data.
- Check DB role privileges.
- Check backup age and restore evidence.